

A Unique Quartic Transition, Möbius Rigidity, and Two Radial Turning Points

for Real-Order Double Polylogarithms

Research continuation prepared for
Vladimir Reshetnikov's ProveIt project

October 10, 2026

Abstract. For $F_{a,b}(z) = \sum_{n>m\geq 1} z^n / (n^a m^b)$, write the upper angular zero near the origin as $\cos \theta = \rho \eta_{a,b}(\rho)$ and expand $\eta = \mu + K\rho^2 + Q\rho^4 + R\rho^6 + \dots$. The audited ProveIt manuscript establishes a unique quadratic threshold $a = A(b)$, but leaves the number of zeros of $q(b) = Q(A(b), b)$ and their higher-order geometry open. We prove, with exact arithmetic certificates, that q has precisely one zero in $0 < b < 1$. Its location is enclosed to 40 decimal places, its derivative is positive, and the sixth-order radial coefficient there is strictly negative. These facts yield a complete local two-parameter bifurcation theorem: an open parameter region has two radial turning points, a minimum followed by a maximum. We also give a fully certified rational example, prove a fourth-root onset law, and classify constant normalized-radius curves for $a \geq 1, b > 0$: only $(a, b) = (1, 1)$ is possible. An all-order coefficient identity and its Möbius transport explain the rigidity and give an order-nine symmetry obstruction at the transition. All analytic arguments are written out; the decisive finite inequalities replay with the Python standard library alone.

Contents

1	Research target and contribution	2
2	The analytic zero equation	2
2.1	Conventions and the local branch	2
2.2	A usable tail bound	3
3	All-order identities and Möbius transport	4
3.1	A coefficient-extraction identity	4
3.2	The sixth-order radial coefficient	4
3.3	An exact Möbius identity	5
4	The quadratic threshold and its scope	6
5	Exactly one quartic transition	7
6	The degenerate point is nondegenerate at the next order	9
7	The full local two-parameter bifurcation	10
7.1	An explicit asymptotic two-turning-point family	11
7.2	The fourth-root onset law	11
8	A certified rational witness	12
9	Rigidity of the constant curve and an order-nine defect	13
10	Audit, proposed manuscript updates, and limitations	14
11	Further research questions	14
A	The finite verification argument in reproducible form	15
A.1	Outward arithmetic and elementary functions	15
A.2	Third-order derivatives along the threshold	15
A.3	Why the mesh is exhaustive	16
A.4	Replay and artifacts	16
B	Proof dependencies and what has actually been established	17

1 Research target and contribution

This article continues the real-order and angular-zero part of *Polylogarithms and their Arithmetic Bridges*. The source audit is pinned to the ProveIt repository state 0707425e155707e53d2892e30c72e54d06c00e5f. The two directly relevant sections are 05-real-local-radius.tex and 05-real-turning.tex [1, 2]. The former proves the sign of the first radial coefficient and the existence of a unique fractional threshold. The latter proves a maximum at inner order 1/2, a minimum at inner order 999/1000, and the existence of an intervening quartic zero. Its closing paragraph explicitly leaves the number of quartic zeros and the higher-order radial behavior at those zeros open.

We resolve these finite-coefficient questions globally on $0 < b < 1$ and then use the answer to obtain new analytic consequences for the actual polylogarithm, rather than only for a Taylor polynomial. The main advances are: a unique quartic transition; its nondegenerate sixth-order behavior; a two-turning-point parameter region and a rational witness; an onset exponent 1/4 at the transition; and a constant-curve rigidity theorem. The all-order identities in Section 3 are additional exact functional and coefficient relations, not conjectured numerical period reductions.

The proofs use standard analytic implicit-function arguments, elementary one-variable bifurcation analysis, and explicit finite sums of real powers. The global quartic sign theorem is computer-assisted. Its numerical inputs are rational boxes whose validity is checked independently by integer arithmetic. The box generator's floating-point root finder is not part of the proof. The available literature on harmonic sums and polylogarithms provides a broader algebraic setting [5], while formal Lagrange inversion supplies a standard coefficient-extraction tool [4]. We do not assert that a targeted literature search establishes worldwide novelty.

2 The analytic zero equation

2.1 Conventions and the local branch

For $a \geq 0$, $b \geq 0$, define

$$H_{n-1}^{(b)} = \sum_{m=1}^{n-1} m^{-b}, \quad p_n(a, b) = \frac{H_{n-1}^{(b)}}{n^a}, \quad F_{a,b}(z) = \sum_{n=2}^{\infty} p_n(a, b) z^n. \quad (2.1)$$

For positive orders this is $\text{Li}_{a,b}(z, 1)$ in the convention $n > m$. We work initially in $|z| < 1$; no branch-cut convention is needed. The value $b = 0$ will only be used to close an interval in the finite-coefficient argument. Since $0 < p_n \leq n - 1$, the series and all its z derivatives converge normally on compact subdisks. Dependence on a, b is real analytic: on any compact complex neighborhood of a real parameter pair, the coefficients have a uniform polynomial growth bound.

Let $z = \rho e^{i\theta}$, with $0 < \rho < 1$ and $0 < \theta < \pi$, and set

$$t = \rho^2, \quad y = \frac{\cos \theta}{\rho}. \quad (2.2)$$

The manuscript proves a unique upper-arc angular zero throughout its stated real-order domain [3]. Our main local arguments need only the branch constructed below; the explicit witness will also have its own certified existence strip.

Theorem 2.1 (Exact angular generating equation). *Put*

$$P_k(y) = \sum_{j=0}^{k+1} (-1)^j \binom{k+1}{j} (2y)^{k+1-j} p_{k+2+j} \quad (k \geq 0). \quad (2.3)$$

Then, near $t = 0$,

$$\Psi(t, y) := \sum_{k \geq 0} P_k(y) t^k = \frac{\operatorname{Im} F_{a,b}(\rho e^{i\theta})}{\rho^3 \sin \theta} \quad \text{when } t = \rho^2, \quad \cos \theta = \rho y. \quad (2.4)$$

There is a unique real-analytic solution $y = \Phi(a, b, t)$ near $t = 0$, with

$$\Phi(a, b, 0) = \mu(a, b) := \frac{p_3}{2p_2}. \quad (2.5)$$

Consequently $\eta_{a,b}(\rho) = \Phi(a, b, \rho^2)$ is even and real analytic near zero.

Proof. The identity $\sin(n\theta)/\sin \theta = U_{n-1}(\cos \theta)$ and the elementary expansion

$$U_{n-1}(x) = \sum_{j=0}^{\lfloor (n-1)/2 \rfloor} (-1)^j \binom{n-1-j}{j} (2x)^{n-1-2j}$$

give the right side of (2.4). A term with indices n, j has radial power $\rho^{2n-4-2j} = t^{n-2-j}$. Regrouping at $k = n - 2 - j$ gives (2.3). The only apparent negative power, from $n = 2$, cancels because $U_1(\rho y) = 2\rho y$. For $|y| \leq M$,

$$|P_k(y)| \leq (2k+3)(1+2M)^{k+1}. \quad (2.6)$$

Thus regrouping is justified when $|t|(1+2M) < 1$. At $t = 0$ the equation is $2p_2 y - p_3 = 0$, and its y derivative is $2p_2 > 0$. The analytic implicit-function theorem proves all remaining claims. $\square$

The same equation can be derived without trigonometry. If z, w are the roots of $X^2 - 2tyX + t = 0$, then

$$\Psi(t, y) = \frac{F(z) - F(w)}{t(z - w)}.$$

This is an identity of symmetric power series, with the removable factors understood at the origin. For the actual angular arc, $w = \bar{z}$. This formulation makes the algebraic symmetry in later sections natural.

2.2 A usable tail bound

Let $L = 1 + 2M$, $q = L|t| < 1$, and truncate after $P_N t^N$. Summing (2.6) gives

$$\left| \Psi(t, y) - \sum_{k=0}^N P_k(y) t^k \right| \leq Lq^{N+1} \left(\frac{2N+5}{1-q} + \frac{2q}{(1-q)^2} \right). \quad (2.7)$$

This bound does not exploit cancellation and therefore remains valid for all nonnegative orders. With $n = N + 1$, a bound for the t derivative of the tail is

$$L^2 q^{n-1} \left\{ \frac{2n^2 + 3n}{1-q} + \frac{(4n+3)q}{(1-q)^2} + \frac{2q(1+q)}{(1-q)^3} \right\}. \quad (2.8)$$

Both inequalities follow by summing a geometric series and its first two derivatives. For real $1/4 \leq y \leq 3/4$, the complex y disk of radius $1/4$ lies in $|y| \leq 1$. Cauchy's estimate therefore bounds the y derivative of the tail by four times (2.7) with $M = 1$. These explicit estimates are used in the rational witness, so its conclusions concern the infinite function.

3 All-order identities and Möbius transport

3.1 A coefficient-extraction identity

Write

$$\Phi(a, b, t) = \mu + \sum_{N \geq 1} c_N t^N. \quad (3.1)$$

The first terms c_1, c_2, c_3 will be denoted K, Q, R .

Proposition 3.1 (All-order reversion formula). *For every $N \geq 1$,*

$$c_N = \sum_{m=1}^N \frac{(-1)^m}{m(2p_2)^m} [u^{m-1}] \sum_{\substack{k_1 + \dots + k_m = N \\ k_1, \dots, k_m \geq 1}} \prod_{r=1}^m P_{k_r}(\mu + u). \quad (3.2)$$

In particular, c_N depends only on $p_2, \dots, p_{2N+3}$.

Proof. Set $v = \Phi - \mu$. The equation is $v = -(2p_2)^{-1} \sum_{k \geq 1} P_k(\mu + v)t^k$. Introduce an auxiliary multiplier s on the right side and apply the formal identity $[s^m]v = m^{-1}[u^{m-1}]H(t, u)^m$ to $v = sH(t, v)$. For completeness, this identity follows by substituting $s = u/H(t, u)$ in formal coefficient extraction, or by differentiating powers and comparing their residues; it is the standard Lagrange inversion formula [4]. Each factor of H is divisible by t , so after extracting t^N only $m \leq N$ survive. Setting $s = 1$ is thus legitimate in the t -adic topology, and gives (3.2). Formula (2.3) gives the stated finite dependence. Analytic convergence follows independently from Theorem 2.1. $\square$

3.2 The sixth-order radial coefficient

Let $A(y) = P_1(y)$, $B(y) = P_2(y)$, $C(y) = P_3(y)$. Explicitly,

$$A(y) = 4p_3y^2 - 4p_4y + p_5, \quad (3.3)$$

$$B(y) = 8p_4y^3 - 12p_5y^2 + 6p_6y - p_7, \quad (3.4)$$

$$C(y) = 16p_5y^4 - 32p_6y^3 + 24p_7y^2 - 8p_8y + p_9. \quad (3.5)$$

Substitution into $\Psi(t, \Phi(t)) = 0$ gives the exact identities

$$K = -\frac{A(\mu)}{2p_2}, \quad (3.6)$$

$$Q = -\frac{A'(\mu)K + B(\mu)}{2p_2}, \quad (3.7)$$

$$R = -\frac{A'(\mu)Q + 4p_3K^2 + B'(\mu)K + C(\mu)}{2p_2}. \quad (3.8)$$

The formulas for K, Q agree with the audited manuscript. The coefficient R is the next obstruction needed at its unresolved transition. At a simultaneous zero of K, Q it simplifies to

$$R = -\frac{C(\mu)}{2p_2}. \quad (3.9)$$

These expressions are exact in arbitrary real orders; integer relations or numerical fitting are not involved.

For efficient verification let $r_n = p_n/p_2 = (2/n)^a H_{n-1}^{(b)}$ and $s = r_3$. Then $\mu = s/2$ and

$$K = \frac{2sr_4 - s^3 - r_5}{2}, \quad (3.10)$$

$$Q = -2(s^2 - r_4)K - \frac{r_4s^3 - 3r_5s^2 + 3r_6s - r_7}{2}. \quad (3.11)$$

The corresponding ratio formula for R is

$$R = -2(s^2 - r_4)Q - 2sK^2 - 3(r_4s^2 - 2r_5s + r_6)K - \frac{r_5s^4 - 4r_6s^3 + 6r_7s^2 - 4r_8s + r_9}{2}. \quad (3.12)$$

Avoiding repeated interval differentiation of p_2^{-1} substantially reduces enclosure widths.

3.3 An exact Möbius identity

For a real constant c , define the involution and the local coordinate

$$T_c(z) = -\frac{z}{1-2cz}, \quad w = \frac{z}{1-cz}, \quad z = \frac{w}{1+cw}. \quad (3.13)$$

The involution sends w to $-w$. Write $G_c(w) = F(w/(1+cw)) = \sum_{n \geq 2} d_n(c)w^n$.

Theorem 3.2 (Transport of the angular coefficients). *Let $F(z) = \sum_{n \geq 2} p_n z^n$ be holomorphic near zero, with P_k defined by (2.3), the following are identities of convergent local power series:*

$$d_{2j+3}(c) = \sum_{k=0}^j (-1)^{k+1} \binom{j+1}{k+1} c^{2(j-k)} P_k(c), \quad (3.14)$$

$$G_c(w) - G_c(-w) = -\frac{2w^3}{(1-c^2w^2)^2} \Psi\left(-\frac{w^2}{1-c^2w^2}, c\right). \quad (3.15)$$

Proof. Take $w = iv$ for small real v . Then $z = (cv^2 + iv)/(1 + c^2v^2)$, so $t = |z|^2 = v^2/(1 + c^2v^2)$ and $\operatorname{Re} z = c|z|^2$. Equation (2.4) yields

$$\operatorname{Im} G_c(iv) = \frac{v^3}{(1 + c^2v^2)^2} \Psi\left(\frac{v^2}{1 + c^2v^2}, c\right).$$

For real coefficients, twice this imaginary part equals $(G_c(iv) - G_c(-iv))/i$. Replacing iv by w gives (3.15), first on an accumulating set and then as a holomorphic identity. Both sides are polynomial-linear in each p_n at any fixed degree, so the same identity holds for complex coefficients. Expanding $(1 + c^2v^2)^{-k-2}$ and comparing coefficients gives (3.14). Alternatively the latter follows directly from $d_n(c) = \sum_{m=2}^n \binom{n-1}{m-1} (-c)^{n-m} p_m$. $\square$

Corollary 3.3 (Structure of constant normalized-radius curves). *Suppose F has real coefficients, $p_2 > 0$, and its local angular branch has $\eta(\rho) \equiv c$. This happens if and only if*

$$F(T_c(z)) = F(z), \quad \text{equivalently} \quad F(z) = \mathcal{H}\left(\frac{z^2}{(1-cz)^2}\right) \quad (3.16)$$

for a holomorphic germ $\mathcal{H}$ with real coefficients and $\mathcal{H}(0) = 0$. Necessarily $c = p_3/(2p_2)$. If $\mathcal{H}(u) = \sum_{j \geq 1} h_j u^j$, its coefficient identities are

$$p_n = \sum_{j=1}^{\lfloor n/2 \rfloor} h_j \binom{n-1}{2j-1} c^{n-2j} \quad (n \geq 2). \quad (3.17)$$

Proof. Constancy means $\Psi(t, c) = 0$ identically. The invertible substitution in (3.15) makes this equivalent to G_c being even. An even holomorphic germ has the form $\mathcal{H}(w^2)$, and the converse is immediate. The coefficient of z^3 determines c , and expansion of $(1 - cz)^{-2j}$ proves (3.17). $\square$

4 The quadratic threshold and its scope

We recall the manuscript's quadratic-threshold result, including the inequalities needed later for rigidity. A compact proof is included to make the dependency explicit; this result is credited to the audited manuscript, not presented as a new contribution [1].

Theorem 4.1 (Established quadratic classification). *For $a \geq 1$, $b > 0$:*

- (i) $K(a, b) < 0$ for every $a \geq 2$;
- (ii) $K(1, b)$ has the sign of $1 - b$;
- (iii) if $b \geq 1$ and $a > 1$, then $K(a, b) < 0$;
- (iv) for each $0 < b < 1$ there is exactly one $A(b) \in (1, 2)$ with $K(A(b), b) = 0$. Below this threshold $K > 0$, and above it $K < 0$.

The threshold extends analytically to a neighborhood of each endpoint of $[0, 1]$, with $A(1) = 1$, and $K_a(A(b), b) < 0$.

Proof. Put $x = 2^{-b}$, $y = 3^{-b}$, $U = 1 + x$, $V = 1 + x + y$, $W = 1 + x + y + x^2$. Direct substitution in (3.6) gives

$$K = -\frac{1}{2}(2/5)^a G_a, \quad G_a = W + U^3(20/27)^a - 2UV(5/6)^a. \quad (4.1)$$

Here the subscript in G_a is an order parameter, not a derivative. Since $x^2 \leq y$ and $0 \leq x \leq 1$, $U^2/V \leq 4/3$. The rational logarithm bounds

$$\log(27/20) < \frac{7273}{24000}, \quad \log(6/5) > \frac{9}{50}$$

therefore imply

$$\frac{U^2}{2V} \frac{\log(27/20)}{\log(6/5)} < \frac{7273}{6480} < \frac{9}{8} \leq (9/8)^a.$$

This is exactly the inequality $\partial_a G_a > 0$ for $a \geq 1$. The displayed logarithm estimates follow from the alternating expansion of $\log(1 + u)$ with its next-term bound.

For $a = 2$, use $y \leq x^{3/2}$ and set $x = t^2$. The coefficient of y in G_2 is negative, and substitution gives the lower bound

$$G_2 \geq \frac{800t^6 - 2025t^5 + 1833t^4 - 567t^3 - 192t^2 + 233}{1458}.$$

Its degree-six Bernstein coefficients on $[0, 1]$ are

$$\left(\frac{233}{1458}, \frac{233}{1458}, \frac{367}{2430}, \frac{665}{5832}, \frac{55}{486}, \frac{95}{1458}, \frac{41}{729} \right).$$

They are all positive. Thus $G_2 > 0$, and monotonicity gives assertion (i).

At $a = 1$,

$$27G_1 = 20x^3 + 42x^2 - 3x + 2 - (45x + 18)y. \quad (4.2)$$

For $b > 1$, set $t = \sqrt{2x} < 1$. The bound $y \leq t^3/3$ gives $G_1 \geq (1 - t)P_5(t)$, where

$$P_5(t) = \frac{-5t^5 + 10t^4 - 11t^3 + t^2 + 4t + 4}{54}.$$

Its Bernstein coefficients are $2/27, 4/45, 19/180, 14/135, 1/10, 1/18$, so $G_1 > 0$. For $0 < b < 1$, write $p = \log_2 3$. Since $11/7 < p < 2$, Taylor's integral formula at $x = 1/2$ gives

$$x^p \geq \frac{1}{3} + \frac{22}{21}(x - 1/2) + \frac{22}{49}(x - 1/2)^2.$$

Indeed $p(p-1)x^{p-2} > 44/49$ on $1/2 < x < 1$. Using the negative coefficient of y in (4.2) yields

$$G_1 \leq -\frac{(2x-1)(10x^2-337x+334)}{2646} < 0.$$

The quadratic in parentheses decreases on $[1/2, 1]$ and equals 7 at one. Direct substitution gives $G_1 = 0$ at $b = 1$. These facts prove (ii)–(iv), using $\partial_a G_a > 0$. At a threshold, (4.1) gives $K_a < 0$. At $b = 0$, $G_1 = -2/27 < 0$ and $G_2 > 0$, so the same root persists; at $b = 1$ the root is $a = 1$. The implicit-function theorem proves the claimed endpoint extensions. $\square$

5 Exactly one quartic transition

Define on the closed interval $[0, 1]$

$$q(b) = Q(A(b), b). \quad (5.1)$$

The endpoint extensions in Theorem 4.1 make this a real-analytic function near every point of the interval.

Theorem 5.1 (Global quartic sign classification). *There is exactly one $b_{\dagger} \in (0, 1)$ with $q(b_{\dagger}) = 0$. Moreover,*

$$q(b) < 0 \quad (0 < b < b_{\dagger}), \quad q(b) > 0 \quad (b_{\dagger} < b < 1), \quad q'(b_{\dagger}) > 0. \quad (5.2)$$

The zero and its corresponding outer order $a_{\dagger} = A(b_{\dagger})$ satisfy

$$\begin{aligned} 0.9974937898734204210746055930669544902305 &< b_{\dagger}, \\ b_{\dagger} &< 0.9974937898734204210746055930669544902306, \end{aligned} \quad (5.3)$$

$$\begin{aligned} 1.0014301809614951381293517293887015374543 &< a_{\dagger}, \\ a_{\dagger} &< 1.0014301809614951381293517293887015374544. \end{aligned} \quad (5.4)$$

The inequalities are certified enclosures, not rounded root estimates.

The finite certificate underlying the global part of the proof is the following stronger pair of derivative inequalities.

Lemma 5.2 (Certified derivative separation). *The exact replay proves*

$$q'(b) > 10^{-6} \quad (0 \leq b \leq 9/10), \quad q''(b) < -\frac{29}{10^6} \quad (9/10 \leq b \leq 1). \quad (5.5)$$

It also proves $q(9/10) < 0 < q(999/1000)$, while $q(1) = 0$ exactly.

Proof and finite verification specification. The computational claim is reduced to finitely many explicit rational inequalities as follows. Partition $[0, 9/10]$ into 2048 equal cells and $[9/10, 1]$ into 256 equal cells. At every endpoint and midpoint, provide a rational interval $[a_-, a_+]$ for $A(b)$ and verify $K(a_-, b) > 0 > K(a_+, b)$. There are 4609 distinct such intervals. For a cell $[u, v]$ form the rectangle

$$\mathcal{B} = [a_-(v), a_+(u)] \times [u, v].$$

The replay verifies $K_a < 0$ and $K_b < 0$ throughout this rectangle. The signs at the two opposite corners, together with $K_b < 0$, are therefore barriers for the entire graph of A on the cell. No unverified interpolation of the graph is used.

All partial derivatives through degree three are computed by automatic differentiation of finite exponential sums, using exact outward dyadic intervals. It is convenient to replace Q by

$$Q_0 = -\frac{1}{2}(r_4 s^3 - 3r_5 s^2 + 3r_6 s - r_7), \quad s = r_3. \quad (5.6)$$

Since $Q = Q_0 - 2(s^2 - r_4)K$, the restrictions of Q and Q_0 to the threshold, and all their total b derivatives there, agree exactly. Implicit differentiation gives

$$\begin{aligned} A' &= -K_b/K_a, \\ A'' &= -(K_{bb} + 2K_{ab}A' + K_{aa}(A')^2)/K_a, \\ q' &= (Q_0)_b + (Q_0)_a A', \\ q'' &= (Q_0)_{bb} + 2(Q_0)_{ab}A' + (Q_0)_{aa}(A')^2 + (Q_0)_a A''. \end{aligned} \quad (5.7)$$

A further differentiation gives q''' , with its explicit formula in Appendix A.

Let $m = (u + v)/2$ and $h = (v - u)/2$. On a first-part cell use the rigorous centered bound

$$q'(b) \geq \inf q'(m) - h \sup_{[u,v]} |q''|.$$

On a second-part cell use

$$q''(b) \leq \sup q''(m) + h \sup_{[u,v]} |q'''|.$$

The first lower bounds all exceed 10^{-6} ; the negatives of the second upper bounds all exceed $29/10^6$. The exact minimum margins are recorded as integer numerators over 2^{256} in `global_verified.json`. Their decimal diagnostics are $1.1282650208766 \cdot 10^{-6}$ and $2.9857032148394 \cdot 10^{-5}$, respectively. Only the integer comparisons are used in the proof.

For the endpoint signs the same arithmetic gives $q(9/10) \approx -1.86117461963559 \cdot 10^{-7}$ and $2.92779196 \cdot 10^{-11} < q(999/1000) < 2.92779197 \cdot 10^{-11}$. At $(a, b) = (1, 1)$, direct rational substitution in the finite coefficient formulas gives $K = Q = 0$, hence $q(1) = 0$. The programs `verify_global.py` and `verify_local.py` check all the stated inequalities. Elementary-function tails, arithmetic rules, input boxes, full-cover checks, and executable replay instructions are included in the package and described in Appendix A. This is a finite exact proof certificate, not a sampling argument. $\square$

Proof of Theorem 5.1. By Lemma 5.2, q is increasing on $[0, 9/10]$ and is negative at its right endpoint. Thus it has no zero there. On $[9/10, 1]$ it is strictly concave. The opposite signs at $9/10$ and $999/1000$ give an interior zero, and $q(1) = 0$ gives a second zero at the endpoint. A strictly concave function cannot have three distinct zeros: at the middle one it would lie strictly above the chord between the outer two. Thus there is exactly one interior zero. Strict concavity also makes q positive between that zero and one, and its derivative at the interior zero is positive. The preceding signs determine the negative side. The exact localization is proved independently in the next section; its unique local zero must be this globally unique zero. $\square$

Corollary 5.3 (Complete quartic signs at the quadratic threshold). *For every fixed $0 < b < b_\dagger$, decreasing a slightly below $A(b)$ creates a unique small-radius strict maximum. For every fixed $b_\dagger < b < 1$, increasing a slightly above $A(b)$ creates a unique small-radius strict minimum. The assertions concern a sufficiently small radius interval, allowed to depend on b .*

Proof. With $U = \partial_t \Phi$, at $(A(b), b, 0)$ one has $U = 0$, $U_t = 2q(b) \neq 0$, and $U_a = K_a < 0$. The analytic implicit-function theorem produces one zero of U in t . Its side, derivative sign, and extremum type follow from the displayed signs. Since $\partial_\rho \eta = 2\rho U$, the same sign changes hold in the physical radius. This generalizes the two separately certified inner orders in the earlier manuscript. $\square$

6 The degenerate point is nondegenerate at the next order

Set

$$R_{\dagger} = R(a_{\dagger}, b_{\dagger}), \quad \mu_{\dagger} = \mu(a_{\dagger}, b_{\dagger}), \quad J_{\dagger} = \det \frac{\partial(K, Q)}{\partial(a, b)}(a_{\dagger}, b_{\dagger}).$$

Theorem 6.1 (Exact nondegeneracy certificate). *At the point in (5.3)–(5.4),*

$$-3.368744536 \cdot 10^{-11} < R_{\dagger} < -3.368744535 \cdot 10^{-11}, \quad (6.1)$$

$$-8.315139844 \cdot 10^{-10} < J_{\dagger} < -8.315139843 \cdot 10^{-10}, \quad (6.2)$$

$$4.870284716 \cdot 10^{-8} < q'(b_{\dagger}) < 4.870284717 \cdot 10^{-8}, \quad (6.3)$$

$$-0.017073210967 < K_a(a_{\dagger}, b_{\dagger}) < -0.017073210965. \quad (6.4)$$

In particular, $(a, b) \mapsto (K, Q)$ is a real-analytic local coordinate system at this point. The radial branch satisfies

$$\eta_{a_{\dagger}, b_{\dagger}}(\rho) = \mu_{\dagger} + R_{\dagger} \rho^6 + O(\rho^8), \quad (6.5)$$

and is strictly decreasing for sufficiently small positive ρ .

Proof. Here is the existence certificate, not just a residual estimate. Let X be the rational rectangle in (5.3)–(5.4), in (a, b) order, and let c be its exact rational midpoint. Put $f = (K, Q)$. Evaluate $Df(c)$ by intervals and take the exact rational inverse C of its entrywise midpoint. Interval evaluation on the whole rectangle then gives

$$H = I - C Df(X), \quad f\text{-iteration image } \mathcal{N} = c - Cf(c) + H(X - c).$$

The verifier proves $\mathcal{N} \subset \text{int } X$ and

$$\|H\|_{\infty} < 3 \cdot 10^{-32} < 1. \quad (6.6)$$

Indeed its diagnostic upper bound is $2.6668469993303 \cdot 10^{-32}$; the actual comparison is rational. For $T(x) = x - Cf(x)$, the integral mean-value formula shows that $T(X) \subseteq \mathcal{N}$. The same derivative enclosure makes T a contraction on X . Iteration is a Cauchy sequence in the complete closed rectangle and converges to a unique fixed point. Since C is invertible, this fixed point has $K = Q = 0$.

The exact inverse matrix, image intervals, and derivative norm are in `local_verified.json`. Direct interval substitution into (3.12) and the derivatives of (3.10)–(3.11) proves (6.1)–(6.4). The identity

$$q'(b) = \frac{K_a Q_b - K_b Q_a}{K_a} \Big|_{(A(b), b)}$$

provides an additional consistent route to the sign in (6.3). Analytic inversion follows from $J_{\dagger} \neq 0$. Finally, (6.5) follows from $K = Q = 0$ and (3.8); its derivative is $6R_{\dagger} \rho^5 + O(\rho^7) < 0$ for small positive radius. $\square$

Remark 6.2 (Why a high-precision residual was not enough). The determinant has magnitude about $8.3 \cdot 10^{-10}$, so the two coefficient equations are nearly tangent in the original parameters. The sixth-order coefficient is smaller still. A small residual alone would not establish either existence or the required signs. The contraction and full-rectangle derivative bounds address both issues.

7 The full local two-parameter bifurcation

By Theorem 6.1, use the exact coefficient coordinates

$$\kappa = K(a, b), \quad \lambda = Q(a, b) \quad (7.1)$$

near $(a_{\dagger}, b_{\dagger})$. Write $\Phi(\kappa, \lambda, t)$ for the analytic radial branch in these coordinates and put $U = \partial_t \Phi$. The coefficient definitions give

$$U(\kappa, \lambda, t) = \kappa + 2\lambda t + 3R(\kappa, \lambda)t^2 + O(t^3), \quad R(0, 0) = R_{\dagger} < 0. \quad (7.2)$$

In particular $U(\kappa, \lambda, 0) = \kappa$ and $U_t(\kappa, \lambda, 0) = 2\lambda$ exactly.

Theorem 7.1 (Complete local turning-point classification). *There exist $t_0 > 0$, a parameter neighborhood $\mathcal{V}$ of $(0, 0)$, and a real-analytic function φ near zero such that*

$$\varphi(0) = \varphi'(0) = 0, \quad \varphi(\lambda) = \frac{\lambda^2}{3R_{\dagger}} + O(\lambda^3). \quad (7.3)$$

Within $\mathcal{V}$, the following list gives every critical point of $\eta(\rho)$ in $0 < \rho < \sqrt{t_0}$.

Parameter condition	Critical-point behavior
$\lambda \leq 0, \kappa \leq 0$	No critical point.
$\kappa > 0$ (either sign of λ)	Exactly one, a strict maximum.
$\lambda > 0, \kappa < \varphi(\lambda)$	No critical point.
$\lambda > 0, \kappa = \varphi(\lambda)$	One stationary point, not an extremum.
$\lambda > 0, \varphi(\lambda) < \kappa < 0$	Exactly two: a strict minimum, then a strict maximum.
$\lambda > 0, \kappa = 0$	Exactly one positive-radius critical point, a strict maximum.

On the fold $\kappa = \varphi(\lambda)$, the stationary point has $\eta' = \eta'' = 0$ and $\eta''' < 0$. The two-critical-point region is open and nonempty in the original (a, b) plane.

Proof. Since $U_{tt}(0, 0, 0) = 6R_{\dagger} < 0$, choose a sufficiently small closed parameter and t neighborhood on which $U_{tt} < 0$ and $U_{\kappa} > 0$. Choose $t_0 > 0$ inside it. At the base parameters, $U(t_0) < 0$ and $U_t(t_0) < 0$; by shrinking the parameter neighborhood these inequalities hold uniformly. Thus U is strictly concave as a function of t , with a negative value at the upper boundary.

If $\lambda \leq 0$, then $U_t(0) = 2\lambda \leq 0$, so U is strictly decreasing for positive t . Its initial value κ immediately gives the first two cases. If $\lambda > 0$, then $U_t(0) > 0 > U_t(t_0)$, so strict concavity gives one interior maximum of U , at a point $T(\kappa, \lambda)$. The implicit-function theorem applied to $U_t = 0$ constructs T analytically near the origin, even if its analytic extension is negative when $\lambda < 0$. Differentiation at the origin gives

$$T_{\kappa}(0, 0) = 0, \quad T_{\lambda}(0, 0) = -\frac{1}{3R_{\dagger}}.$$

Set $H(\kappa, \lambda) = U(\kappa, \lambda, T(\kappa, \lambda))$. Then $H_{\kappa}(0, 0) = 1$, so $H = 0$ is an analytic curve $\kappa = \varphi(\lambda)$. Along this curve,

$$T = -\frac{\lambda}{3R_{\dagger}} + O(\lambda^2), \quad 0 = H = \varphi(\lambda) - \frac{\lambda^2}{3R_{\dagger}} + O(\lambda^3).$$

This proves (7.3); in particular $\varphi(\lambda) < 0$ for small nonzero λ . Since $H_\kappa > 0$ in a smaller neighborhood, the sign of the maximum of U is the sign of $\kappa - \varphi(\lambda)$. Together with $U(0) = \kappa$ and $U(t_0) < 0$, this gives every remaining row of the table. Strict concavity precludes any additional root in this interval.

At a simple root, $\eta_{\rho\rho} = 4tU_t$. Thus the upward crossing is a strict minimum and the downward crossing is a strict maximum. At a double root of U , one has $U = U_t = 0$ and

$$\eta_{\rho\rho\rho} = 8\rho^3 U_{tt} < 0.$$

There is no change of sign in η' there, hence no local extremum. Finally, analytic inversion of (7.1) maps the open nonempty wedge $\varphi(\lambda) < \kappa < 0$, $\lambda > 0$, to an open nonempty region in the original parameter plane. $\square$

Remark 7.2 (Terminology and scope). The discriminant curve here is a fold tangent to the boundary $\kappa = 0$ at $\lambda = 0$. The theorem does not identify it with a Whitney cusp. It classifies every critical point in a fixed small radius interval for nearby parameters, not every critical point on the whole unit disk. In particular it does not refute the manuscript's separate integer-outer-order conjecture for $a \geq 2$.

7.1 An explicit asymptotic two-turning-point family

Corollary 7.3 (Separated small-radius extrema). *There is an analytic order pair $(a(\varepsilon), b(\varepsilon))$ for small ε with*

$$K = 6R_\dagger \varepsilon^2, \quad Q = -\frac{9}{2}R_\dagger \varepsilon. \quad (7.4)$$

For every sufficiently small $\varepsilon > 0$, its normalized-radius curve has exactly two critical points in the local interval of Theorem 7.1, with

$$\rho_{\min} = \sqrt{\varepsilon}(1 + O(\varepsilon)), \quad \rho_{\max} = \sqrt{2\varepsilon}(1 + O(\varepsilon)). \quad (7.5)$$

Proof. Analytic inversion of $(a, b) \mapsto (K, Q)$ defines the family. Substitute $t = \varepsilon s$ into (7.2). Uniformly for bounded s ,

$$\varepsilon^{-2}U = 3R_\dagger(s-1)(s-2) + O(\varepsilon).$$

At each of $s = 1, 2$ the limiting root is simple. The implicit-function theorem gives roots $s = 1 + O(\varepsilon)$ and $s = 2 + O(\varepsilon)$, with the stated types. The complete local classification excludes additional local critical points. $\square$

7.2 The fourth-root onset law

Corollary 7.4 (Fixed-inner-order onset at the transition). *Fix $b = b_\dagger$ and let $a \uparrow a_\dagger$ from below. There is exactly one small-radius critical point, a strict maximum, and*

$$\rho_{\max}(a, b_\dagger) = \left(\frac{K_a(a_\dagger, b_\dagger)}{3R_\dagger} \right)^{1/4} (a_\dagger - a)^{1/4} + O((a_\dagger - a)^{3/4}). \quad (7.6)$$

For $a > a_\dagger$ sufficiently close there is no small-radius critical point. The ratio inside the fourth root is positive.

Proof. Let $h = a_\dagger - a > 0$. Then $\kappa = -K_a(a_\dagger, b_\dagger)h + O(h^2) > 0$ and $\lambda = O(h)$. Set $t = \sqrt{h}s$ in (7.2); after division by h the limiting equation is $-K_a(a_\dagger, b_\dagger) + 3R_\dagger s^2 = 0$. Its positive root is simple. Applying the implicit-function theorem in $\sqrt{h}$ and then taking a positive square root of t proves (7.6). Uniqueness and maximality follow from Theorem 7.1. For $a > a_\dagger$, $\kappa < 0$ and $\lambda = O(|a - a_\dagger|)$. If $\lambda \leq 0$ there is no root by the first row of the table. If $\lambda > 0$, then $\varphi(\lambda) = O((a - a_\dagger)^2)$, whereas κ is negative of first order. Thus $\kappa < \varphi(\lambda)$ for sufficiently small positive $a - a_\dagger$, and there is again no root. $\square$

The previously proved maxima and minima at nonzero $q(b)$ have a square-root radius law. The exponent $1/4$ is a genuine change of local scaling, forced by the vanishing quartic coefficient and the nonzero sixth-order coefficient; it is not a fit to a numerical curve.

8 A certified rational witness

The preceding existence theorem has unspecified neighborhood sizes. The following result gives a concrete order pair and a fully explicit radius interval. Each terminating decimal in the statement denotes the exact rational number, not an approximation to an implicitly defined pair.

Theorem 8.1 (A rational curve with at least two turning points). *Take*

$$\begin{aligned} a_0 &= 1.0014300032325009280102804945924913210012973562600080008779598088, \\ b_0 &= 0.99749410117972379489824653149307623117370381382325680750587755176. \end{aligned}$$

The branch $\eta_{a_0, b_0}(\rho)$ has a strict local minimum in

$$\frac{1}{\sqrt{20000}} < \rho < \sqrt{\frac{3}{20000}}$$

and a strict local maximum at a larger radius in

$$\sqrt{\frac{3}{20000}} < \rho < \frac{1}{\sqrt{4000}}.$$

In particular, at least two positive-radius turning points occur below $\rho = 1/\sqrt{4000}$.

Proof. Use (2.4) with $N = 16$, so only $p_2, \dots, p_{35}$ are required. The exact interval replay constructs an η strip of half-width 10^{-8} centered at a rational enclosure midpoint of $\mu(a_0, b_0)$. Throughout $0 \leq t \leq 1/4000$, the left boundary has $\Psi < 0$, the right boundary has $\Psi > 0$, and

$$0.99900 < \Psi_y < 0.99906 \tag{8.1}$$

throughout the strip. These inequalities use the infinite-series tails (2.7)–(2.8), together with the Cauchy bound for the y derivative. There is therefore a unique analytic root $y = \Phi(t)$ on this entire explicit interval. It is the branch from Theorem 2.1.

At each of three rational values of t , the replay encloses the root in an interval of half-width 10^{-42} and uses $\Phi'(t) = -\Psi_t/\Psi_y$. It certifies the following inequalities:

t	Exact enclosure for $\Phi'(t)$
$1/20000$	$-7.59 \cdot 10^{-19} < \Phi'(t) < -7.56 \cdot 10^{-19}$
$3/20000$	$2.52 \cdot 10^{-19} < \Phi'(t) < 2.54 \cdot 10^{-19}$
$1/4000$	$-7.59 \cdot 10^{-19} < \Phi'(t) < -7.56 \cdot 10^{-19}$

The verifier is `verify_witness.py`; exact endpoints, all tail bounds, the root strip, and the rational order pair are retained in `witness_verified.json`. Since $\partial_\rho \eta = 2\rho\Phi'(t)$, the signs in the physical radius are negative, positive, negative. Analyticity and the nonzero endpoint derivatives imply a strict local minimum between the first two samples and a strict local maximum between the last two. This proves the assertion for the exact infinite polylogarithm. $\square$

Remark 8.2 (What this certificate does not count). The witness proves at least two extrema in the displayed interval. It does not claim that this particular rational pair has exactly two critical points on $0 < \rho \leq 1$. The local classification proves an exact count in its own uniform small neighborhood; the explicit witness is independently certified and does not assume that its parameters lie in an unspecified neighborhood from an asymptotic theorem.

The witness was found by taking $\varepsilon = 10^{-4}$ in (7.4), solving numerically for an order pair, and rounding it to the displayed rational values. This history is not used in the proof: the replay begins with those rational values and proves the three signs afresh. The tiny derivative sizes also explain why an ordinary double-precision plot would not be reliable evidence.

9 Rigidity of the constant curve and an order-nine defect

Theorem 9.1 (Constant-curve classification). *For $a \geq 1$ and $b > 0$, the normalized-radius branch is constant on a neighborhood of $\rho = 0$ if and only if $(a, b) = (1, 1)$. In that case*

$$F_{1,1}(z) = \frac{1}{2} \log^2(1 - z), \quad \eta_{1,1}(\rho) = \frac{1}{2} \quad (0 < \rho \leq 1). \quad (9.1)$$

Equivalently, within this order range, a Möbius symmetry $F_{a,b}(T_c(z)) = F_{a,b}(z)$ can occur only for $(a, b, c) = (1, 1, 1/2)$.

Proof. A constant curve has $K = Q = R = 0$. The quadratic classification leaves only $(a, b) = (1, 1)$ or a pair $(A(b), b)$ with $0 < b < 1$. In the latter case the quartic classification forces $b = b_{\dagger}$. But Theorem 6.1 gives $R(a_{\dagger}, b_{\dagger}) < 0$, a contradiction.

For $(1, 1)$, termwise differentiation of (2.1) gives

$$F'_{1,1}(z) = -\frac{\log(1 - z)}{1 - z}.$$

The zero initial value proves the first identity in (9.1). On the upper disk, the imaginary part of the logarithm is nonzero. Thus $\text{Im} \log^2(1 - z) = 0$ exactly when $|1 - z| = 1$, which is $\cos \theta = \rho/2$. This proves the second identity, including the boundary by continuation at this zero. The symmetry characterization follows from Corollary 3.3. $\square$

Corollary 9.2 (A quantified obstruction to the next flat curve). *At $(a_{\dagger}, b_{\dagger})$, set $c = \mu_{\dagger}$ and $w = z/(1 - cz)$. Then*

$$F_{a_{\dagger}, b_{\dagger}}(z) - F_{a_{\dagger}, b_{\dagger}}(T_c(z)) = -4 \cdot 2^{-a_{\dagger}} R_{\dagger} w^9 + O(w^{11}). \quad (9.2)$$

The leading coefficient is strictly positive. The transformed function is even through degree eight but fails to be even at degree nine.

Proof. At $c = \mu$, $P_0(c) = 0$. Since $K = Q = 0$, (3.6)–(3.7) also give $P_1(c) = P_2(c) = 0$. The transport identity then gives $d_3 = d_5 = d_7 = 0$ and $d_9 = P_3(c) = -2p_2 R_{\dagger}$. Taking the odd part of G_c proves (9.2), and $p_2 = 2^{-a_{\dagger}} > 0$, $R_{\dagger} < 0$ determine the sign. $\square$

This supplies a structural explanation for the nearly flat transition: it satisfies the first three odd symmetry conditions but not the fourth. The functional relation (3.15) makes the explanation exact, rather than merely an analogy between a Taylor expansion and a geometric picture.

10 Audit, proposed manuscript updates, and limitations

The formulas for K and Q in the audited local-radius and turning sections agree with the independent derivations here. We found no incorrect formula in those inspected portions. The appropriate integration is therefore a theorem-status update, not a claim that the previously cautious statements were errors.

The closing paragraph of `05-real-turning.tex` can now be updated in three precise ways. The number of zeros of q in $(0, 1)$ is one; the displayed decimal location can be replaced by the certified intervals (5.3)–(5.4); and the higher-order behavior can cite the negative sixth-order coefficient, the two-turning-point region, and the fourth-root onset law. The earlier separate maximum and minimum theorems should remain credited as predecessors and useful special cases of Corollary 5.3.

The source’s warning that a nonzero local coefficient does not determine whole-disk monotonicity remains important. Our two-turning-point theorem strengthens that warning. It does not prove the full-radius conjecture for integer outer orders $a \geq 2$, classify all fractional curves on the unit disk, or extend the constant-curve theorem to $0 < a < 1$. Nothing here proves the manuscript’s separate S_6 or S_8 arithmetic candidates, numerical period independence, or a minimal-depth claim.

All finite sign decisions in the main certificates use only Python integers and rational input values. The optional symbolic and numerical regressions are distinguished from the proof. There is no Lean formalization, and successful replay does not replace independent review of the analytic reductions and interval implementation. The trust base is explicit: the written analytic arguments, Python integer arithmetic, the interval code, and the supplied rational boxes.

The package is designed for placement under `Analysis/Polylogarithms/docs/reports/radial-boundary-bifurcation/`. It contains a namespaced manuscript excerpt, bibliography additions, a narrow proposed status patch, provenance, and replay instructions. No repository files were modified as part of preparing this article.

11 Further research questions

1. **Whole-disk turning-point counts.** Determine whether the explicit rational witness has exactly two critical points on $0 < \rho \leq 1$. More generally, classify the continuation of the local two-turning-point region up to the unit circle. A local fold does not by itself exclude further critical points at larger radii.
2. **The integer-outer-order conjecture.** Prove or disprove strict decrease of $\eta_{a,b}$ on the whole interval for $a \geq 2$, $b > 0$, and the proposed $a = 1$ alternatives. The present counterexamples all have fractional outer order strictly between one and two, so the integer conjecture survives unchanged.
3. **Constant curves below outer order one.** Extend Theorem 9.1 to $0 < a < 1$. The Möbius normal form reduces the question to an infinite, triangular family of exact coefficient identities. The first three obstructions already suffice for $a \geq 1$, but a corresponding classification of $K = 0$ below one is not supplied here.
4. **An analytic proof of quartic uniqueness.** Replace the 2304-cell certificate by a direct inequality for $q'(b)$ on $[0, 9/10]$ and for $q''(b)$ on $[9/10, 1]$, or find a simpler single transformed quantity with a fixed derivative sign. The present certificate suggests strict margins but is not such a symbolic proof.
5. **Higher contact and multiple polylogarithms.** For deeper real-order polylogarithms, derive the analogue of (3.15) and investigate how many independent odd symmetry conditions can vanish. Determine whether higher-dimensional order families produce radial

onset exponents beyond $1/4$.

6. **Sharp quantitative neighborhoods.** Give explicit t_0 and parameter radii for every row of Theorem 7.1. An interval Taylor model adapted to (K, Q) coordinates should be more efficient than a raw rectangle in (a, b) because the coefficient map is poorly conditioned.
7. **Formal and cross-system verification.** Formalize the interval arithmetic and the centered implicit-derivative bounds in a proof assistant, or independently replay the analytic inequalities with a separately implemented interval system. This would reduce implementation risk without conflating finite checking with formalization of the entire analytic argument.

A The finite verification argument in reproducible form

A.1 Outward arithmetic and elementary functions

Every interval is represented by integer endpoints $[L, U]$ denoting $[L2^{-256}, U2^{-256}]$. Addition and negation are exact on this grid. For multiplication, the four integer endpoint products are computed; the least is divided by 2^{256} with floor rounding and the greatest with ceiling rounding. A positive reciprocal interval is obtained from $2^{512}/U$ and $2^{512}/L$ with outward rounding. Negative reciprocals are reduced by sign reversal. Division by an interval containing zero is rejected. The resulting rules are inclusion-preserving by elementary monotonicity, so an induction over an expression proves that its computed interval contains its exact range.

For $2 \leq n \leq 9$, use

$$\log n = 2 \sum_{j=0}^{N-1} \frac{z^{2j+1}}{2j+1} + E_N, \quad 0 < E_N < \frac{2z^{2N+1}}{(2N+1)(1-z^2)}, \quad z = \frac{n-1}{n+1}, \quad N = 512. \quad (\text{A.1})$$

The witness also needs $n \leq 35$. For $n > 9$, write $n = 2^k m$ with $1 \leq m < 2$, use the same formula for $\log m$ with $N = 128$, and add $k \log 2$. Here $z = (n - 2^k)/(n + 2^k) < 1/3$. For $0 \leq x \leq 8$, set $y = x/16$ and use

$$e^y = \sum_{j=0}^{64} \frac{y^j}{j!} + E, \quad 0 \leq E \leq \frac{y^{65}}{65!} \frac{1}{1 - y/66}. \quad (\text{A.2})$$

Four interval squarings enclose e^x , and reciprocation encloses e^{-x} . All exponential arguments in the certificates lie in the allowed interval. The logarithm and exponential tail bounds are positive geometric majorants; no floating-point library supplies a proof bound.

A.2 Third-order derivatives along the threshold

The implementation stores the ten Taylor coefficients with multi-indices (i, j) satisfying $i + j \leq 3$. Entries are divided derivatives $\partial_a^i \partial_b^j / (i!j!)$, so multiplication is ordinary truncated convolution. Inversion is obtained recursively from the identity $XX^{-1} = 1$. The derivatives of each normalized coefficient can also be checked directly from

$$r_n = (2/n)^a \sum_{m < n} m^{-b}, \quad \partial_a^i \partial_b^j r_n = (-\log(n/2))^i (2/n)^a \sum_{m < n} (-\log m)^j m^{-b}.$$

This gives a short independent specification of every jet input.

Let $v = A'$, $w = A''$, as in (5.7), and define

$$\mathcal{C}(X) = X_{bbb} + 3X_{abb}v + 3X_{aab}v^2 + X_{aaa}v^3 + 3(X_{ab} + X_{aa}v)w. \quad (\text{A.3})$$

Then

$$A''' = -\frac{\mathcal{C}(K)}{K_a}, \quad q''' = \mathcal{C}(Q_0) + (Q_0)_a A'''.$$

These are obtained by differentiating $K(A(b), b) = 0$ and $q(b) = Q_0(A(b), b)$ three times. They are used only after a strict negative enclosure for K_a has been established.

A.3 Why the mesh is exhaustive

The input file `global_mesh.json` is a list of rational proposals. The verifier first checks that the first endpoint is zero, the last is one, adjacent endpoints agree exactly, all cell lengths are positive, and each listed midpoint is the exact arithmetic midpoint. It checks that every cell lies in the correct side of $9/10$. It then verifies both signs for every root bracket, checks the barrier derivatives on every cell rectangle, and checks that the midpoint bracket lies within its rectangle. Thus missing intervals, unverified root guesses, and unjustified graph interpolation are not accepted.

The centered derivative step is essential. A natural interval evaluation of q' on a moderately wide rectangle loses too much correlation between a and b . Instead the exact midpoint value is combined with a bound on one higher total derivative along the implicitly defined graph. The mean-value theorem supplies the rigorous error. The same method one order higher proves concavity near $b = 1$.

A.4 Replay and artifacts

From the package root, the decisive checks are

```
python code/verify_global.py
python code/verify_local.py
python code/verify_witness.py
```

They require only the Python standard library. They must not be run with `-O`; optimized execution is explicitly rejected because the verifiers use assertions for proof obligations. The optional check

```
python code/verify_symbolic.py
```

requires SymPy and verifies the coefficient identities at finite orders, the sixth-order formula, and the Bernstein expansion. It is a regression check, not an extrapolation argument. The universal identities are proved in the text. The optional generator `make_mesh.py` requires mpmath; regenerating boxes is unnecessary to verify the delivered certificate.

Artifact	Role
<code>global_mesh.json</code>	Rational graph-cover proposals.
<code>global_verified.json</code>	All 2304 strict cell margins and coverage receipt.
<code>local_verified.json</code>	Contraction, location, Jacobian and sixth-order signs.
<code>witness_verified.json</code>	Rational orders, analytic tails, root strip and three signs.
<code>symbolic_verified.json</code>	Optional exact symbolic regression receipt.
<code>integration/</code>	Namespaced excerpt, bibliography and proposed status replacement.
<code>provenance/</code>	Audited repository paths, blob identifiers and build environment.

The final archive also includes editable LaTeX, a compiled PDF, the source code, and a checksum manifest. Receipts record successful replay for this package; they do not automatically validate subsequently edited files.

B Proof dependencies and what has actually been established

The logic is deliberately separated into four layers. First, the generating equation, reversion formula, Möbius transport, and local bifurcation arguments are ordinary analytic or formal proofs. Second, the known quadratic classification supplies the globally defined threshold $A(b)$; its proof is reproduced and credited. Third, finitely many exact inequalities prove derivative separation, local contraction, and an explicit witness. Fourth, the analytic arguments transfer those inequalities to uniqueness, local extrema, scaling laws, and rigidity for the infinite function.

The new global statement is global in the inner-order parameter $0 < b < 1$ *along the quadratic threshold*. The new exact turning-point count is local in disk radius and in the two order parameters. The concrete rational witness supplies an explicit small radius interval, but only an at-least-two count. These three scopes are different and should not be merged during manuscript integration.

The central resolved question can be summarized without any numerical notation: along the unique quadratic threshold in the strip $1 < a < 2$, $0 < b < 1$, the quartic coefficient changes sign exactly once; at that change the sixth-order coefficient is negative. This finite coefficient fact forces an open two-extremum region and rules out a second constant normalized-radius curve for every $a \geq 1$, $b > 0$.

References

- [1] ProveIt Contributors, *A proved local part of the normalized-radius conjecture*, `Analysis/Polylogarithms/docs/manuscript/chapters/05-real-local-radius.tex`, Git blob 30e3e31944be5bc0d03a8485386dd1608c075507, audited October 10, 2026. Repository: <https://github.com/VladimirReshetnikov/ProveIt>.
- [2] ProveIt Contributors, *A proved failure of fractional global monotonicity*, `Analysis/Polylogarithms/docs/manuscript/chapters/05-real-turning.tex`, Git blob 49f0336656b6aa893e5ba6e904fb561c3a941cfb, audited October 10, 2026, at repository commit 0707425e155707e53d2892e30c72e54d06c00e5f.
- [3] ProveIt Contributors, *Cartesian motion of the complete real-order zero arc*, `Analysis/Polylogarithms/docs/manuscript/chapters/05-real-cartesian-motion.tex`, Git blob 0deb9bb54adaef8783d4e0e7ce43d426643450a0, audited October 10, 2026.
- [4] Ira M. Gessel, *Lagrange inversion*, Journal of Combinatorial Theory, Series A **144** (2016), 212–249. <https://doi.org/10.1016/j.jcta.2016.06.018>.
- [5] Jakob Ablinger, Johannes Blümlein, and Carsten Schneider, *Harmonic sums and polylogarithms generated by cyclotomic polynomials*, Journal of Mathematical Physics **52** (2011), 102301. <https://arxiv.org/abs/1105.6063>.